

Question Paper

Exam Date & Time: 29-Nov-2025 (02:00 PM - 05:00 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MANIPAL INSTITUTE OF TECHNOLOGY - BENGALURU
SEVENTH SEMESTER B.TECH. DEGREE EXAMINATIONS - NOVEMBER 2025

Distributed Systems [CSE_3251]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) Identify the major design challenges in building distributed systems. (2)
A) Course Outcome: "CO1" / Cognitive Level: "Application"
B) Show the client-server and peer-to-peer architectural styles. (3)
Course Outcome: "CO1" / Cognitive Level: "Application"
C) Illustrate an example architecture of a distributed system such as Google File System (GFS). (5)
Course Outcome: "CO1" / Cognitive Level: "Analysis"
- 2) Show the layers involved in communication in distributed systems. (2)
A) Course Outcome: "CO2" / Cognitive Level: "Application"
B) Demonstrate the role of stubs in RPC implementation. (3)
Course Outcome: "CO2" / Cognitive Level: "Application"
C) Outline multicast communication and its applications. (5)
Course Outcome: "CO2" / Cognitive Level: "Analysis"
- 3) Manipulate the following words Names, Identifiers, and Addresses in distributed systems. (2)
A) Course Outcome: "CO3" / Cognitive Level: "Application"
B) Demonstrate the concept of flat naming with an example. (3)
Course Outcome: "CO3" / Cognitive Level: "Application"
C) Illustrate the structured naming and how name resolution is performed. (5)
Course Outcome: "CO3" / Cognitive Level: "Analysis"
- 4) Show concept of clock synchronization? Why is it needed? (2)
A) Course Outcome: "CO4" / Cognitive Level: "Application"

B) Illustrate Lamport's logical clock with an example. (3)

Course Outcome: "CO4" / Cognitive Level: "Analysis"

C) Describe distributed mutual exclusion algorithm. (5)

Course Outcome: "CO4" / Cognitive Level: "Evaluate"

5) Show consistency and replication in distributed systems. (2)

A) Course Outcome: "CO5" / Cognitive Level: "Application"

B) Outline session guarantees used in client-centric consistency. (3)

Course Outcome: "CO5" / Cognitive Level: "Analysis"

C) Compare primary-based and replicated-write consistency protocols. (5)

Course Outcome: "CO5" / Cognitive Level: "Evaluate"

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Event Name	Subject
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058-END SEMSTER-ODD-2025 CSE_3251:Distributed Systems

#	Course Outcome	Description	Marks
1	1	CO 1	10
2	2	CO 2	10
3	3	CO 3	10
4	4	CO 4	10
5	5	CO 5	10
Total			50

* **NA** - CO Not Selected

Event Name	Subject
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058-END SEMSTER-ODD-2025 CSE_3251:Distributed Systems

#	Level	No of Questions	Marks (%)
1	Remember	0	0(0%)
2	Understand	0	0(0%)
3	Application	8	19(38%)
4	Analysis	5	21(42%)
5	Create	2	10(20%)
6	Evaluate	0	0(0%)
Total		15	50 (100%)